

More samples or a higher sampling rate?

Background

Finite sinc reconstruction has truncation error because samples outside the observed window are omitted. Aliasing is different: f_s below the Nyquist requirement makes different continuous signals share samples. A longer observation window does not remove that ambiguity. RMSE is the square root of the mean squared error.

Your task

For $x(t) = \cos(2\pi 3t) + 0.5 \cos(2\pi 13t)$, run four experiments: $(f_s, M) = (20, 20), (20, 80), (40, 20), (40, 80)$. Each uses sample indices $-M, \dots, M$. Reconstruct on the fixed grid and measure RMSE against both the original and the folded reference. Print one result row per experiment and select the row with the smallest original-signal RMSE.

Function contract and assumptions

`run_case(fs, M, t_eval)` returns $(xr, \text{error_original}, \text{error_folded})$. `best_case(cases, t_eval)` returns the zero-based index of the smallest original-signal RMSE, with the earliest row winning a tie. Use `t_eval` as 1001 equally spaced points from -0.25 to 0.25 inclusive. Signal, folding, sinc reconstruction, and RMSE helpers are supplied; complete only the experiment logic.

Example / correctness check

At $f_s=20$, the folded reference is $\cos(2\pi 3t) + 0.5 \cos(2\pi 7t)$. At $f_s=40$, it equals the original. Each case has $2M+1$ samples. Do not assume that increasing M always decreases every finite-window error; select the best case from the computed values.

Exam instructions

Complete only the TODO functions in `template.py` and preserve their signatures. The supplied driver handles printing and plotting. Your functions must work for all valid inputs described above, not just the example. No external files or packages beyond NumPy and Matplotlib are needed. Small floating-point differences are acceptable.

Topic reference: PDF pages 19-22 and 27; finite-window error and experiment comparisons are explained above.